



Solo Security Lab

A Safe Cybersecurity Learning Environment

Version: 1.0.0

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GitHub: github.com/G1H6-bit/solo-security-lab

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1. Project Overview

Solo Security Lab is a self-contained Python desktop application that teaches beginners what real cyberattacks look like in server logs — entirely on their own machine, with zero risk. No virtual machine required. No external network connections. No prior experience needed.

The application spins up a fake web server, fires simulated attacks against it, and shows the results in a live color-coded log viewer — all inside a single window. It transforms abstract attack concepts into something visual and immediately understandable.

2. Project Information

Language	Python 3.x
GUI Library	Tkinter (built-in — no installation required)
Networking	Python built-in http.server module
Category	Cybersecurity / Educational Tool / Desktop Application
Difficulty	Intermediate
Dependencies	None — runs with standard Python 3 only
Platform	Windows / macOS / Linux

3. Architecture — Three Systems in One

When you launch Solo Security Lab, three independent components start simultaneously inside a single window:

Component	Description	Technology
 Fake Web Server	Boots at 127.0.0.1:8080 and writes Apache-style log entries	http.server + threading
 Attack Panel	5 buttons, each firing a different simulated attack at the server	requests + threading

Component	Description	Technology
 Live Log Viewer	Displays every incoming request color-coded by attack type	Tkinter Text widget

4. The 5 Attack Simulations

Each attack button fires a pre-programmed pattern at the local fake server. The log viewer captures and highlights the results in real time:

Attack	What It Simulates	Real-World Equivalent
 Rapid Flood	Sends a burst of rapid HTTP requests	Denial-of-Service (DoS) attempt
 Directory Scan	Probes common URL paths like /admin, /login, /config	Reconnaissance / path traversal
 Fake Scanner	Mimics automated vulnerability scanner user-agents	Nikto / OpenVAS scanner behavior
 Stealth Probe	Slow, spaced-out requests designed to avoid detection	APT low-and-slow attack pattern
 All Combined	Fires all four attacks simultaneously	Coordinated multi-vector attack

5. Key Features

- **Live color-coded log viewer — each attack type has a distinct color**
- Scoreboard tracking the total requests per attack type
- Export full session log as a .txt file for review
- Built-in Help Guide explaining each attack concept
- Runs 100% on your own machine — no internet connection required
- No extra libraries — only standard Python 3
- Safe and ethical — all traffic stays on localhost (127.0.0.1)

6. How to Run

Prerequisites

- Python 3.x installed
- No additional pip packages required

Setup & Launch

```
git clone https://github.com/G1H6-bit/solo-security-lab.git
cd solo-security-lab
python solo-security-lab.py
```

7. Cybersecurity Concepts Demonstrated

This project teaches the following concepts through hands-on simulation:

Concept	How It Is Demonstrated
Denial of Service (DoS)	Rapid Flood attack overwhelms the request handler
Web Reconnaissance	Directory Scan probes for hidden or sensitive paths
Automated Scanning	Fake Scanner mimics known vulnerability scanner signatures
Low-and-Slow Attacks	Stealth Probe spaces requests to evade rate limiting
Log Analysis	Live log viewer shows how defenders read and interpret attack patterns
Multi-Vector Attacks	All Combined demonstrates how attackers use simultaneous methods

8. Why This Project Stands Out

- No other beginner project lets you see DoS, recon, and stealth attacks side by side in real time
- Bridges the gap between theory and practice without needing a full lab environment
- The export feature means you can document and study your own attack sessions
- Shows professional awareness of ethical boundaries — 100% localhost, zero risk

9. What I Learned

- Python multithreading — running a server, attacker, and UI simultaneously
- How real web servers generate Apache-format log entries
- The technical signature of DoS, reconnaissance, and stealth attack patterns
- Building a real-time log viewer with Tkinter's Text widget
- Ethical hacking principles — safe, contained environments for security testing

10. Future Improvements

- Add a Detection Engine that flags suspicious patterns automatically
- Include a quiz mode to test the user's ability to identify attack types
- Add more attack types: SQL injection patterns, XSS attempts, brute force
- Save and replay past attack sessions

11. License & Author

This project is released under the MIT License — open source and free to use.

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